TED (15) 3124 Reg. No………………….

REVISION 2015 Signature……………..

THIRD SEMESTER DIPLOMA EXAMINATION IN WOOD AND PAPER TECHNOLOGY

WOOD CHEMISTRY

Maximum Marks: 100 Time: 3 Hours

PART- A

I. Answer the following questions in one or two sentences. (2X5=10 marks)

1. Define methylated spirit.
2. State gels.
3. State alpha cellulose.
4. State pyroligneous acid.
5. Definetropolenes.

PART- B

II. Answer any five of the following. (5X6=30 marks)

1. Predictthe influence of extractives on wood.
2. Distinguish between saturated and unsaturated compounds.
3. Describe the reactions of cellulose.
4. Describe the analysis of wood.
5. Summarize the uses of radio isotopes.
6. Describe the action of enzymes on wood.
7. Distinguish lyophilic and lyophobic colloids.

PART- C

Answer one full question from each module. (15X4=60 marks)

Module- I

III. 1. Apply IUPAC rules to draw the structure of the following.

i. 4-iodoheptanoic acid ii. 3-methyl -2-pentanol

iii. 2-ethyl heptane iv. 1-chloro-2-ethyl octane 8

2. Describe the structure of glucose and cellulose. 7

OR

IV. 1. Apply IUPAC rules to name the following.

i. $CH_3-CH_2-CH_2-COOH$

ii. $CH_3-CH_2-O-CH_2-CH_3$

iii. $CH_3-CH(OH)-CH_3$

iv. $CH_3-CH_2-CH=CH_2$ 8

2. Explain the isolation of terpenes. 7

Module- II

V. 1. Rewrite the various methods for the preparation of colloids. 8

2. Explain the corrosion control methods. 7

OR

VI. 1. Explain in one's own words the effect of radioactivity. 8

2. Point out the applications of colloids. 7

MODULE- III

VII. 1. Sketch the structure of wood cell walland describe the parts. 10

2. Classify the components of wood. 5

OR

VIII. 1. Explain the extraction of various components in wood. 10

2. Describe steam distillation in wood. 5

MODULE- IV

IX. 1. Discuss the destructive distillation in wood. 9

2. Describe the wood distillation products. 6

OR

X. 1. Demonstrate the determination of moisture content in wood. 9

2. Explain various types of wood hydrolysis. 6